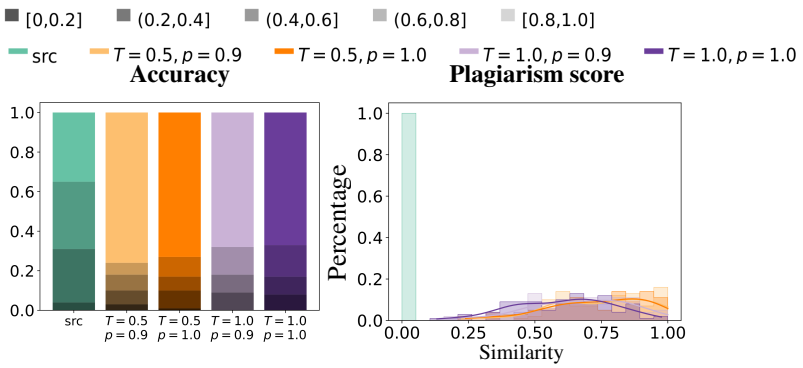

Figure 20: **Model**: GPT-4. **(Left)** Stacked bar charts of the accuracy values achieved by the source solutions as well as GPT-4 generated solutions under various kwargs. **(Right)** Histogram plus kernel density estimation (KDE) plots of the plagiarism scores.

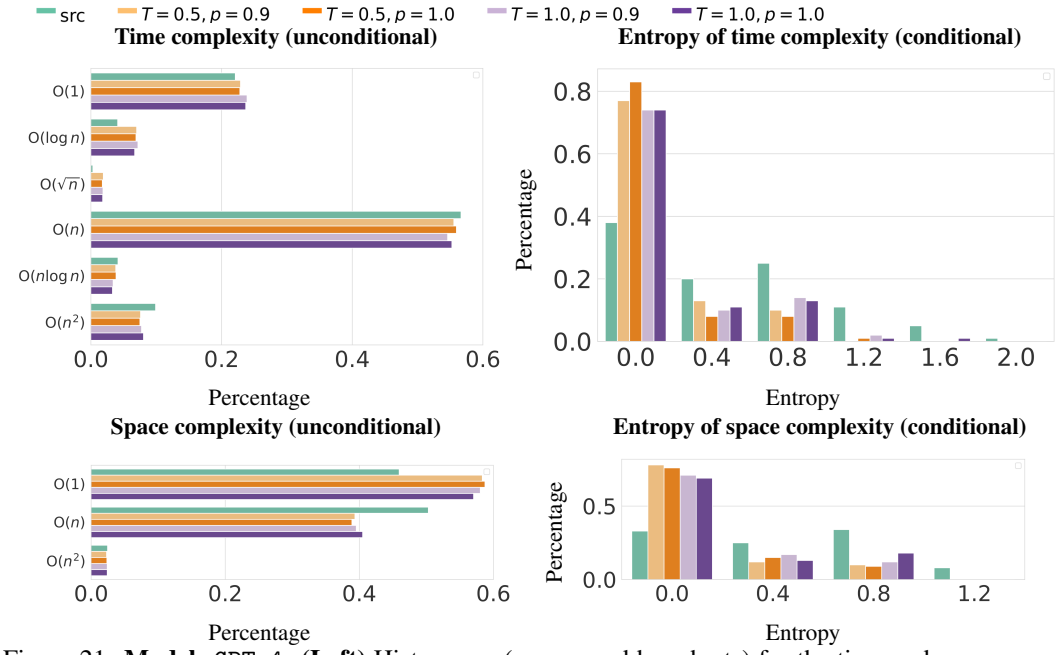

Figure 21: **Model**: GPT-4. **(Left)** Histograms (or grouped bar charts) for the time and space complexity for the unconditional distribution. **(Right)** Histograms for the *entropy* of the time and space complexity for the conditional distribution. The figures suggest that the generated code is more efficient than the source code, while showing a decrease in diversity.

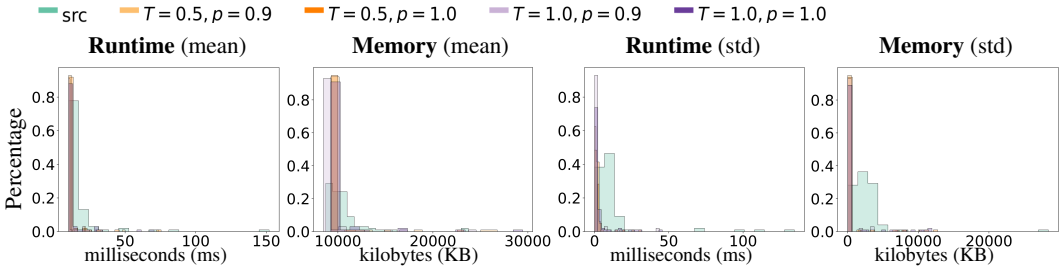

Figure 22: **Model**: GPT-4. **Histograms of the mean and standard deviation (std) of runtime** (in milliseconds) **and memory usage** (in kilobytes). We filtered out some datapoints for better visualization, specifically, runtime above 200 ms, and memory usage above 30,000 KB.

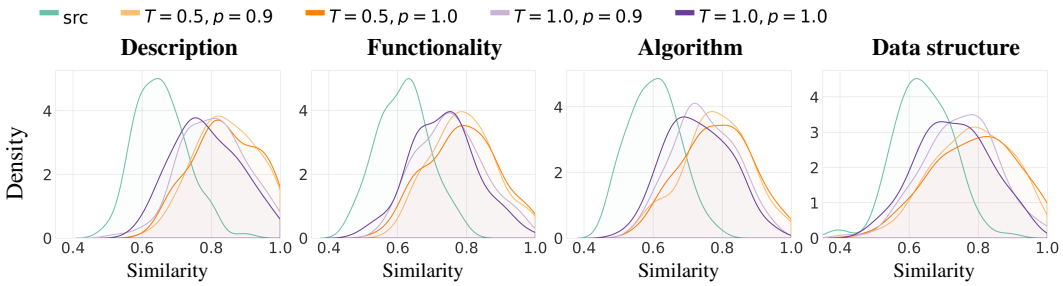

Figure 23: **Model**: `GPT-4`. **Kernel density estimation (KDE) plots of the mean pairwise cosine similarity** for the four attributes (description, functionality, algorithm, and data structure) represented as extracted embeddings of natural language descriptions.

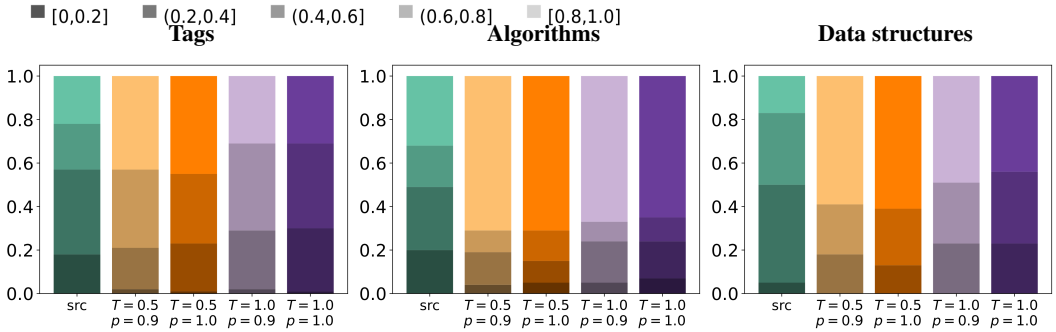

Figure 24: **Model**: `GPT-4`. **Stacked bar charts of the mean pairwise Jaccard index** for the three attributes (tags, algorithms, and data structures) represented as sets of categorical variables.

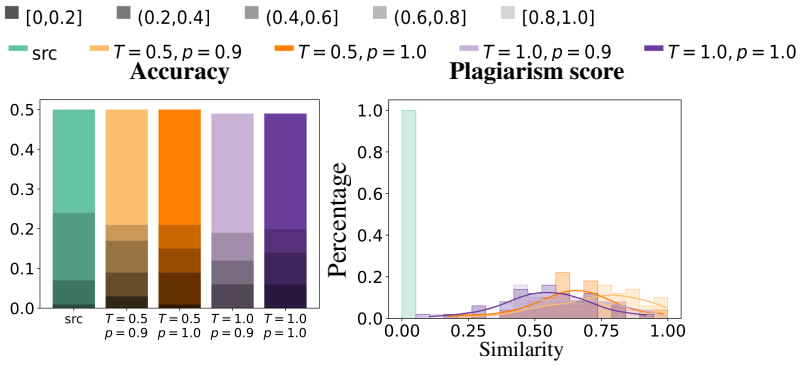

Figure 25: **Model**: `Claude-3-Sonnet`. **(Left)** Stacked bar charts of the accuracy values achieved by the source solutions as well as Claude generated solutions under various kwargs. **(Right)** Histogram plus kernel density estimation (KDE) plots of the plagiarism scores for source solutions and Claude generated solutions.

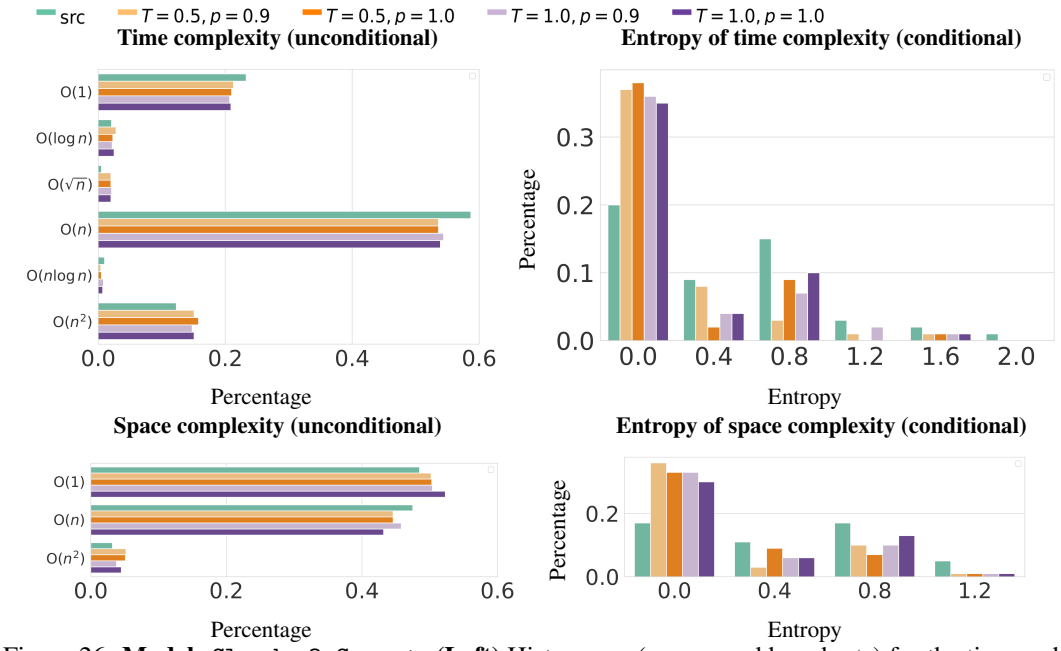

Figure 26: **Model**: `Claude-3-Sonnet`. **(Left)** Histograms (or grouped bar charts) for the time and space complexity for the unconditional distribution. **(Right)** Histograms for the *entropy* of the time and space complexity for the conditional distribution. The figures suggest that the generated code is more efficient than the source code, while showing a decrease in diversity.

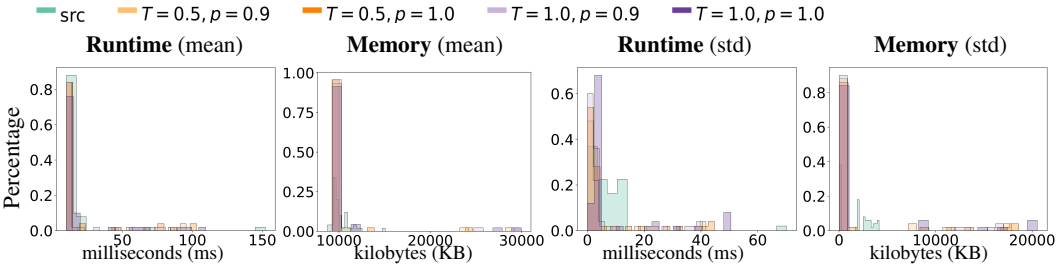

Figure 27: **Model**: `Claude-3-Sonnet`. **Histograms of the mean and standard deviation (std) of runtime** (in milliseconds) **and memory usage** (in kilobytes). We filtered out some datapoints for better visualization, specifically, runtime above 200 ms, and memory usage above 30,000 KB.

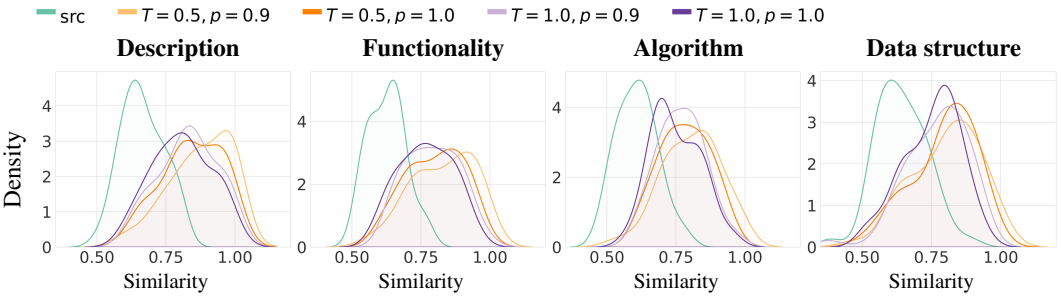

Figure 28: **Model**: `Claude-3-Sonnet`. **Kernel density estimation (KDE) plots of the mean pairwise cosine similarity** for the four attributes (description, functionality, algorithm, and data structure) represented as extracted embeddings of natural language descriptions.